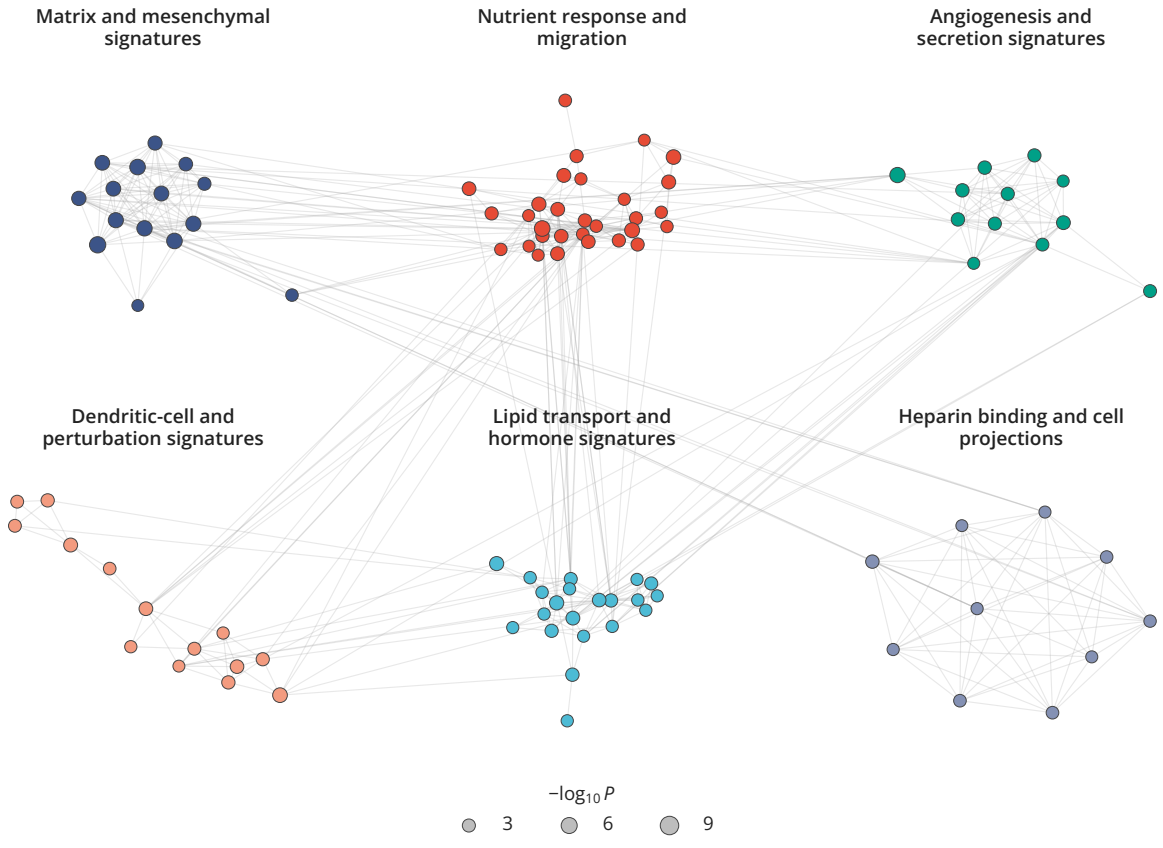
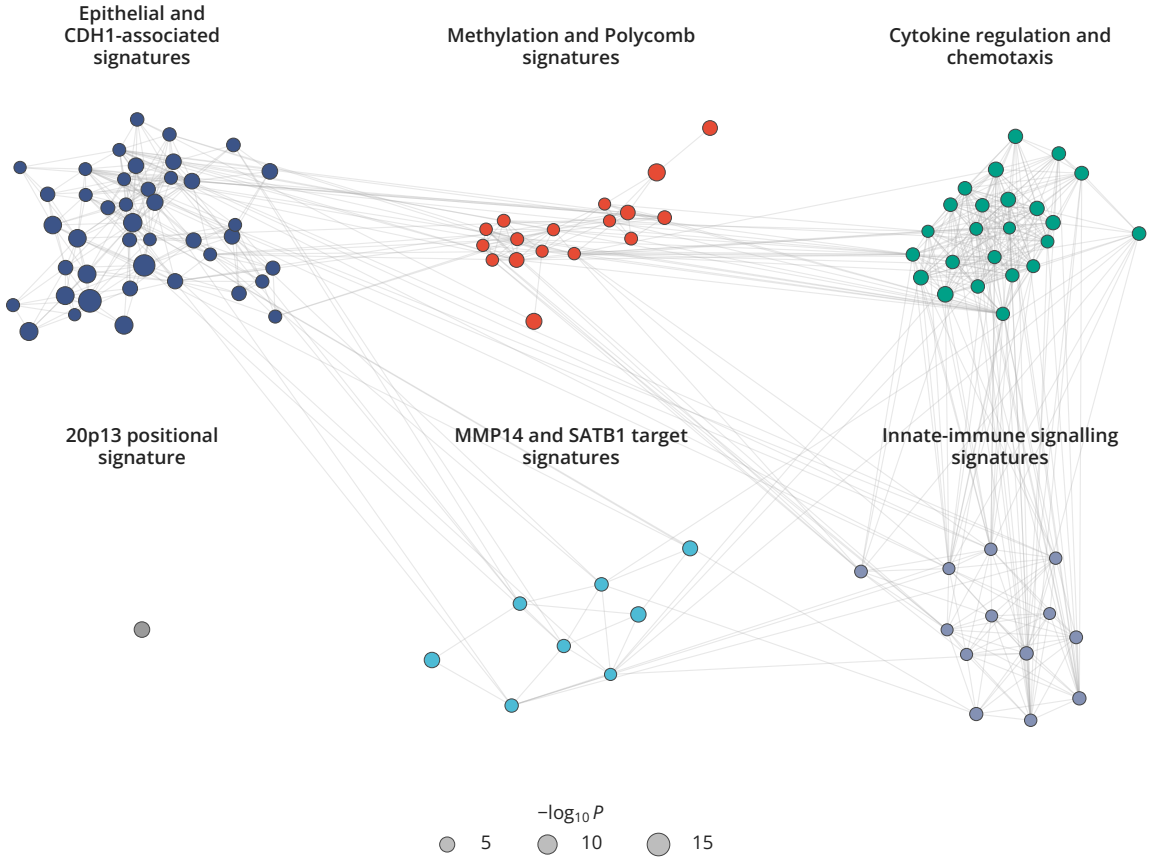
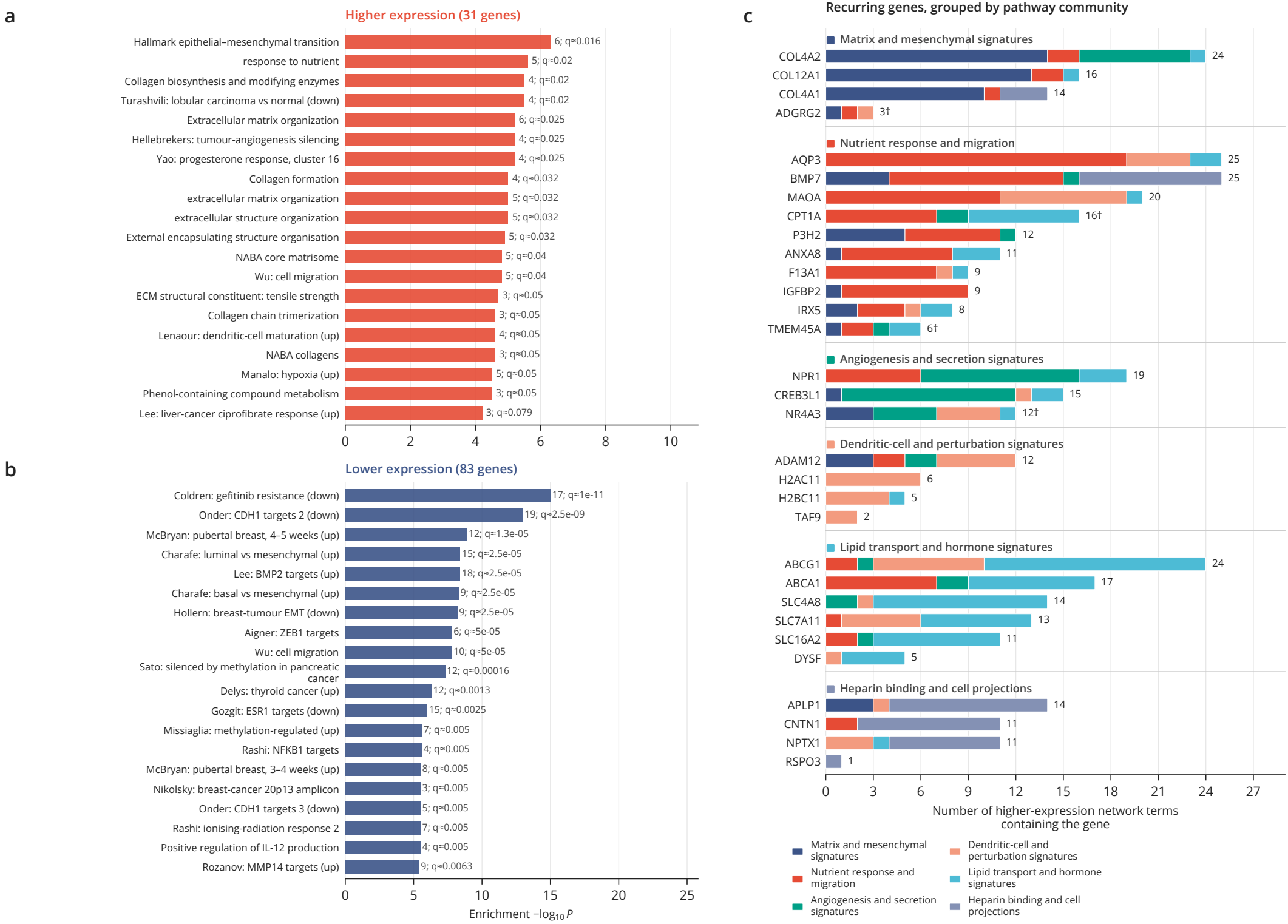


Theme	Terms	Best −log₁₀P	Representative terms	Recurring genes
Matrix and mesenchymal signatures	15	6.3	Hallmark epithelial–mesenchymal... Collagen biosynthesis and...	COL4A2, COL12A1, COL4A1, P3H2
Nutrient response and migration	30	5.6	response to nutrient Wu: cell migration	AQP3, BMP7, MAOA, IGFBP2
Angiogenesis and secretion signatures	12	5.2	Hellebrekers: tumour-angiogenesis... secretion	CREB3L1, NPR1, COL4A2, NR4A3
Dendritic-cell and perturbation signatures	14	4.6	Lenaour: dendritic-cell maturation... Alcoholism	MAOA, ABCG1, H2AC11, ADAM12
Lipid transport and hormone signatures	21	4.1	intracellular lipid transport Gozgit: ESR1 targets (up)	ABCG1, SLC4A8, ABCA1, SLC16A2
Heparin binding and cell projections	10	3.3	heparin binding neuron projection morphogenesis	APLP1, BMP7, CNTN1, NPTX1



Theme	Terms	Best $-\log_{10}P$	Representative terms	Recurring genes
Epithelial and CDH1-associated signatures	42	15.0	Coldren: gefitinib resistance... Onder: CDH1 targets 2 (down)	TACSTD2, LCN2, CLDN7, ST14
Methylation and Polycomb signatures	17	7.3	Sato: silenced by methylation in... Gozgit: ESR1 targets (down)	TACSTD2, DLC1, KDR, BST2
Cytokine regulation and chemotaxis	24	5.5	Positive regulation of IL-12... Tarte: plasma cell vs plasmablast...	SYK, MDK, CCL22, CSF1
20p13 positional signature	1	5.5	Nikolsky: breast-cancer 20p13...	ADISSP, ATRN, CENPB
MMP14 and SATB1 target signatures	8	5.4	Rozanov: MMP14 targets (up) Han: SATB1 targets (up)	CALD1, CXCL1, EFEMP1, MPZL2
Innate-immune signalling signatures	13	3.3	Amoebiasis Chemokine signaling pathway	PLCB1, PLCB4, B2M, CXCL1


a
Higher expression (31 genes)
Hallmark epithelial–mesenchymal transition
6; q≈0.016
response to nutrient
5; q≈0.02
Collagen biosynthesis and modifying enzymes
4; q≈0.02
Turashvili: lobular carcinoma vs normal (down)
4; q≈0.02
Extracellular matrix organization
6; q≈0.025
Hellebrekers: tumour-angiogenesis silencing
4; q≈0.025
Yao: progesterone response, cluster 16
4; q≈0.025
Collagen formation
4; q≈0.032
extracellular matrix organization
5; q≈0.032
extracellular structure organization
5; q≈0.032
External encapsulating structure organisation
5; q≈0.032
NABA core matrisome
5; q≈0.04
Wu: cell migration
5; q≈0.04
ECM structural constituent: tensile strength
3; q≈0.05
Collagen chain trimerization
3; q≈0.05
Lenaour: dendritic-cell maturation (up)
4; q≈0.05
NABA collagens
3; q≈0.05
Manalo: hypoxia (up)
5; q≈0.05
Phenol-containing compound metabolism
3; q≈0.05
Lee: liver-cancer ciprofibrate response (up)
3; q≈0.079
0
2
4
6
8
10
b
Lower expression (83 genes)
Coldren: gefitinib resistance (down)
17; q≈1e-11
Onder: CDH1 targets 2 (down)
19; q≈2.5e-09
McBryan: pubertal breast, 4–5 weeks (up)
12; q≈1.3e-05
Charafe: luminal vs mesenchymal (up)
15; q≈2.5e-05
Lee: BMP2 targets (up)
18; q≈2.5e-05
Charafe: basal vs mesenchymal (up)
9; q≈2.5e-05
Hollern: breast-tumour EMT (down)
9; q≈2.5e-05
Aigner: ZEB1 targets
6; q≈5e-05
Wu: cell migration
10; q≈5e-05
Sato: silenced by methylation in pancreatic cancer
12; q≈0.00016
Delys: thyroid cancer (up)
12; q≈0.0013
Gozgit: ESR1 targets (down)
15; q≈0.0025
Missiaglia: methylation-regulated (up)
7; q≈0.005
Rashi: NFKB1 targets
4; q≈0.005
McBryan: pubertal breast, 3–4 weeks (up)
8; q≈0.005
Nikolsky: breast-cancer 20p13 amplicon
3; q≈0.005
Onder: CDH1 targets 3 (down)
5; q≈0.005
Rashi: ionising-radiation response 2
7; q≈0.005
Positive regulation of IL-12 production
4; q≈0.005
Rozanov: MMP14 targets (up)
9; q≈0.0063
0
5
10
15
20
25
Enrichment −log10 P
c
Recurring genes, grouped by pathway community
Matrix and mesenchymal signatures
COL4A2
24
COL12A1
16
COL4A1
14
ADGRG2
3†
Nutrient response and migration
AQP3
25
BMP7
25
MAOA
20
CPT1A
16†
P3H2
12
ANXA8
11
F13A1
9
IGFBP2
9
IRX5
8
TMEM45A
6†
Angiogenesis and secretion signatures
NPR1
19
CREB3L1
15
NR4A3
12†
Dendritic-cell and perturbation signatures
ADAM12
12
H2AC11
6
H2BC11
5
TAF9
2
Lipid transport and hormone signatures
ABCG1
24
ABCA1
17
SLC4A8
14
SLC7A11
13
SLC16A2
11
DYSF
5
Heparin binding and cell projections
APLP1
14
CNTN1
11
NPTX1
11
RSPO3
1
0
3
6
9
12
15
18
21
24
27
Number of higher-expression network terms containing the gene
Matrix and mesenchymal signatures
Nutrient response and migration
Angiogenesis and secretion signatures
Dendritic-cell and perturbation signatures
Lipid transport and hormone signatures
Heparin binding and cell projections